HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 15 Name: _____ Marks: / 83

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Consider the compound inequality

$$3x - 4 < 5 \leq \frac{2 - 7x}{6} \quad \text{..... (*).}$$

(a) Solve $(*)$.

(b) Write down the greatest integer satisfying $(*)$.

(4 marks) (DSE-2026 PI Q06)

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2. A profit of \$162 is made by selling a ring at a discount of 15% on its marked price. The marked price of the ring is \$270 above its cost. Find the percentage profit. (4 marks) (DSE-2026 PI Q07)

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3. The frequency distribution table and the cumulative frequency distribution table below show the distribution of the weights of 20 apples.

Weight (grams)	Frequency
91 – 100	2
101 – 110	a
111 – 120	6
121 – 130	7
131 – 140	h

Weight less than (grams)	Cumulative frequency
100.5	2
110.5	6
120.5	12
130.5	b
140.5	k

(a) Find a and b.

(b) Find the mean of the distribution.

(5 marks) (DSE-2026 PI Q09)

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4. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find w.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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5. It is given that $f(x)$ partly varies as x^2 and is partly constant. Suppose that $f(4) = 0$ and $f(8) = -9$. Denote the graph of $y = f(x)$ by Γ.

(a) Find the y-intercept of Γ.

(b) Γ cuts the x-axis at the points U and V while Γ cuts the y-axis at the point W. Find the perimeter of $\triangle UVW$.

(6 marks) (DSE-2026 PI Q11)

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6. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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7. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders X, Y and Z. Let r_1 cm, r_2 cm and r_3 cm be the base radii of X, Y and Z respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of Y in terms of π.

(b) A craftsman finds that the height of Y is 9 cm.

(i) Find r_2.

(ii) The craftsman claims that the total surface area of Z is greater than the surface area of the sphere. Do you agree? Explain your answer.

(8 marks) (DSE-2026 PI Q13)

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8. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a.

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer.

(8 marks) (DSE-2026 PI Q14)

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9. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn;

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn.

(4 marks) (DSE-2026 PI Q15)

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10. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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11. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

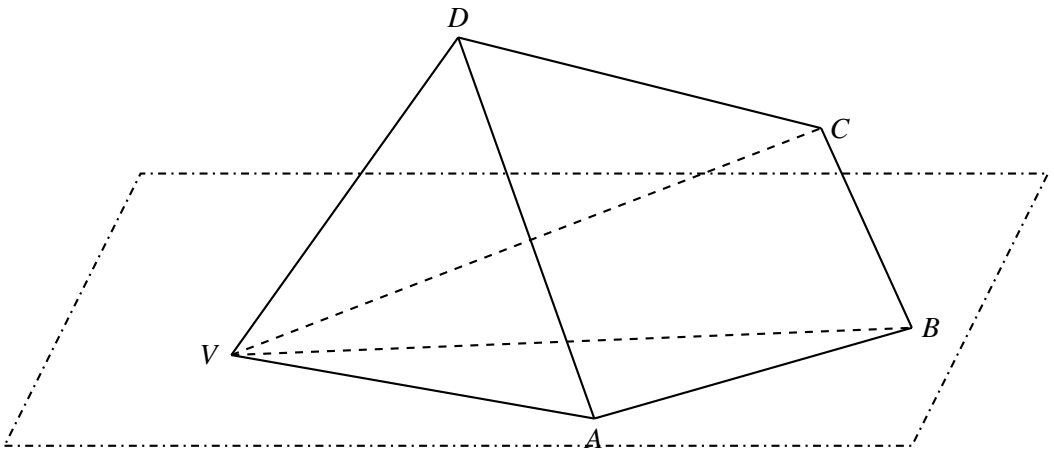


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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12. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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13. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4 - x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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I can finish _____ questions

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